

Boyuan Zhang

Assistant Professor of Computer Science

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EDUCATION

Indiana University Bloomington

Aug. 2021 – Jun. 2026

Ph.D. in Intelligent Systems Engineering

- Advisors: Prof. Fengguang Song and Prof. Dingwen Tao
- Started at Washington State University (2021–2022) and transferred to IU with Prof. Tao's group.

University of Southern California

Jan. 2019 – Dec. 2020

M.S. in Electrical Engineering

Shanghai Jiao Tong University

Sep. 2014 – Jun. 2018

B.Eng. in Information Engineering

RESEARCH INTERESTS

- **High-Performance Computing (HPC):** Design and optimization of scalable systems for scientific and engineering workloads, with a focus on efficiency and parallel performance.
- **Data Compression:** Development of GPU-based lossy and lossless compression algorithms for scientific data and machine learning workloads, emphasizing fidelity, throughput, and scalability.
- **Quantum Computing:** Exploration of high-performance methods for large-scale quantum circuit simulation and quantum system modeling, addressing memory and scalability challenges.
- **Machine Learning Systems:** System-level optimizations for training and inference, including model compression, memory footprint reduction, and GPU-accelerated data pipelines.

PROFESSIONAL EXPERIENCE

University of Kentucky, Assistant Professor of Computer Science

Aug. 2026 – Present

ByteDance, Research Scientist Intern

May 2025 – Aug. 2025

Pacific Northwest National Laboratory, Research Collaborator & Intern

May 2023 – Jun. 2026

Argonne National Laboratory, Research Collaborator

Aug. 2022 – Jun. 2025

Meta, Research Collaborator (Meta Research Award)

Aug. 2022 – Aug. 2024

PUBLICATIONS

[IPDPS'26] Boyuan Zhang, Ding Zhou, Yafan Huang, Shihui Song, Hao Feng, Jinda Jia, Chengming Zhang, and Zhi Zhang. *Near-Zero Cost KV Cache Compression for Large Language Model Inference*. In *Proceedings of the 40th IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, 2026.

[IPDPS'26] Boyuan Zhang, Luanzheng Guo, Jiannan Tian, Jinyang Liu, Daoce Wang, Chengming Zhang, Bo Fang, Fengguang Song, Jan Strube, Nathan R. Tallent, and Dingwen Tao. *Accelerating AI Compression through Lightweight Lossless Encoding and Pipelined Workflows*. In *Proceedings of the 40th IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, 2026.

- [ICS'25] (Best Paper Runner-Up) Boyuan Zhang**, Bo Fang, Fanjiang Ye, Luanzheng Guo, Fengguang Song, Nathan R. Tallent, and Dingwen Tao. *BMQSim: Overcoming Memory Constraints in Quantum Circuit Simulation with a High-Fidelity Compression Framework*. In *Proceedings of the 39th ACM International Conference on Supercomputing (ICS)*, pp. 689–704, 2025.
- [ICS'25] (Best Paper Candidate) Boyuan Zhang**, Yafan Huang, Sheng Di, Fengguang Song, Guanpeng Li, and Franck Cappello. *Pushing the Limits of GPU Lossy Compression: A Hierarchical Delta Approach*. In *Proceedings of the 39th ACM International Conference on Supercomputing (ICS)*, pp. 654–669, 2025.
- [SC'24]** Hao Feng, **Boyuan Zhang**, Fanjiang Ye, Min Si, Ching-Hsiang Chu, Jiannan Tian, Chunxing Yin, Summer Deng, Yuchen Hao, Pavan Balaji, Tong Geng, and Dingwen Tao. *Accelerating Communication in Deep Learning Recommendation Model Training with Dual-Level Adaptive Lossy Compression*. In *SC24: International Conference for High Performance Computing, Networking, Storage and Analysis*, pp. 1–16. IEEE, 2024. (Co-first authors.)
- [HPDC'23]** **Boyuan Zhang**, Jiannan Tian, Sheng Di, Xiaodong Yu, Yunhe Feng, Xin Liang, Dingwen Tao, and Franck Cappello. *FZ-GPU: A Fast and High-Ratio Lossy Compressor for Scientific Computing Applications on GPUs*. In *Proceedings of the 32nd International Symposium on High-Performance Parallel and Distributed Computing (HPDC)*, pp. 129–142, 2023.
- [ICS'23]** **Boyuan Zhang**, Jiannan Tian, Sheng Di, Xiaodong Yu, Martin Swamy, Dingwen Tao, and Franck Cappello. *GPULZ: Optimizing LZSS Lossless Compression for Multi-Byte Data on Modern GPUs*. In *Proceedings of the 37th ACM International Conference on Supercomputing (ICS)*, pp. 348–359, 2023.
- [PPoPP'25 Poster]** **Boyuan Zhang**, Luanzheng Guo, Jiannan Tian, Jinyang Liu, Daoce Wang, Fanjiang Ye, Chengming Zhang, Jan Strube, Nathan R. Tallent, and Dingwen Tao. *High-Performance Visual Semantics Compression for AI-Driven Science*. In *Proceedings of the 30th ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP)*, pp. 557–559, 2025.
- [ICS'26]** Ruoyu Li, Yafan Huang, Longtao Zhang, Zhuoxun Yang, Sheng Di, **Boyuan Zhang**, Jiajun Huang, Jinyang Liu, Jiannan Tian, Guanpeng Li, Fengguang Song, Hanqi Guo, Franck Cappello, and Kai Zhao. *GPZ: GPU-Accelerated Lossy Compressor for Particle Data*. In *Proceedings of the 40th ACM International Conference on Supercomputing (ICS)*, 2026.
- [SC'25]** Daoce Wang, Pascal Grosset, Jesus Pulido, Jiannan Tian, Tushar M. Athawale, Jinda Jia, Baixi Sun, **Boyuan Zhang**, et al. *STZ: A High Quality and High Speed Streaming Lossy Compression Framework for Scientific Data*. In *SC25: International Conference for High Performance Computing, Networking, Storage and Analysis*, IEEE, 2025.
- [ACM Computing Surveys'25]** Sheng Di, Jinyang Liu, Kai Zhao, Xin Liang, Robert Underwood, Zhaorui Zhang, Milan Shah, **Boyuan Zhang**, et al. *A Survey on Error-Bounded Lossy Compression for Scientific Datasets*. *ACM Computing Surveys*, 57(11): 1–38, 2025.
- [ICS'25]** Wenqi Jia, Zhewen Hu, Youyuan Liu, **Boyuan Zhang**, Jinzhen Wang, Jinyang Liu, Wei Niu, et al. *NeurLZ: An Online Neural Learning-Based Method to Enhance Scientific Lossy Compression*. In *Proceedings of the 39th ACM International Conference on Supercomputing (ICS)*, pp. 26–42, 2025.
- [PPoPP'25]** Baixi Sun, Weijin Liu, J. Gregory Pauloski, Jiannan Tian, Jinda Jia, Daoce Wang, **Boyuan Zhang**, et al. *COMPSO: Optimizing Gradient Compression for Distributed Training with Second-Order Optimizers*. In *Proceedings of the 30th ACM SIGPLAN Annual Symposium on Principles and Practice of Parallel Programming (PPoPP)*, pp. 212–224, 2025.
- [FGCS'25]** Franck Cappello, Mario Acosta, Emmanuel Agullo, Hartwig Anzt, Jon Calhoun, Sheng Di, Luc Giraud, **Boyuan Zhang**, et al. *Multifacets of Lossy Compression for Scientific Data in the*

Joint-Laboratory of Extreme Scale Computing. Future Generation Computer Systems, 163: 107323, 2025.

[SC'24] Jinyang Liu, Jiannan Tian, Shixun Wu, Sheng Di, **Boyuan Zhang**, Robert Underwood, Yafan Huang, et al. *CUSZ-i: High-Ratio Scientific Lossy Compression on GPUs with Optimized Multi-Level Interpolation*. In *SC24: International Conference for High Performance Computing, Networking, Storage and Analysis*, pp. 1–15. IEEE, 2024.

[DAC'24] Chenxi Li, **Boyuan Zhang**, Yongqiang Duan, Yang Li, Zuochang Ye, Weifeng Liu, Dingwen Tao, and Zhou Jin. *MASC: A Memory-Efficient Adjoint Sensitivity Analysis through Compression Using Novel Spatiotemporal Prediction*. In *Proceedings of the 61st ACM/IEEE Design Automation Conference (DAC)*, pp. 1–6, 2024.

PRESENTATIONS & TALKS

- 05/2026, paper presentation, “Near-Zero Cost KV Cache Compression for Large Language Model Inference.” IPDPS'26, New Orleans, LA, USA.
- 05/2026, paper presentation, “Accelerating AI Compression through Lightweight Lossless Encoding and Pipelined Workflows.” IPDPS'26, New Orleans, LA, USA.
- 06/2025, paper presentation, “BMQSim: Overcoming Memory Constraints in Quantum Circuit Simulation with a High-Fidelity Compression Framework.” ICS'25, Salt Lake City, UT, USA.
- 06/2025, paper presentation, “Pushing the Limits of GPU Lossy Compression: A Hierarchical Delta Approach.” ICS'25, Salt Lake City, UT, USA.
- 02/2025, poster presentation, “High-Performance Visual Semantics Compression for AI-Driven Science.” PPOPP'25, Las Vegas, NV, USA.
- 06/2023, paper presentation, “FZ-GPU: A Fast and High-Ratio Lossy Compressor for Scientific Computing Applications on GPUs.” HPDC'23, Orlando, FL, USA.
- 06/2023, paper presentation, “GPULZ: Optimizing LZSS Lossless Compression for Multi-Byte Data on Modern GPUs.” ICS'23, Orlando, FL, USA.

TEACHING

- Fall 2026, Instructor. CS 660: “Scaling HPC & ML via Data Reduction” University of Kentucky.
- Spring 2026, Assistant Instructor. ENGR-516: “Engineering Cloud Computing” Indiana University.
- Fall 2024, Assistant Instructor. ENGR-516: “Engineering Cloud Computing” Indiana University.
- Spring 2022, Teaching Assistant. CPTS 360 “Systems Programming” Washington State University.
- Fall 2021, Teaching Assistant. CPTS 360 “Systems Programming” Washington State University.

AWARDS & HONORS

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| ▪ ICS'25 Best Paper Runner-up (BMQSim), Top 3 of 320 submissions | Jun. 2025 |
| ▪ ICS'25 Best Paper Candidate (Aatrox), Top 6 of 320 submissions | Jun. 2025 |
| ▪ Student Travel Grant (\$1,500), HPDC'23 | Jun. 2023 |
| ▪ Graduate Conference Funding (\$3,000), Indiana University Bloomington | Jun. 2022 |

PROFESSIONAL SERVICE

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| ▪ Program Committee (PC) Member, PPOPP'27 | 2026 |
| ▪ Program Committee (PC) Member, IPDPS'27 | 2026 |

▪ Publicity Chair, NPC'27	2026
▪ Web & Publicity Chair, SSDBM'27	2026
▪ Student Volunteer, SC'25, St. Louis, MO, USA	2025
▪ Web Chair, QCCC'24	2024
▪ Student Volunteer, SC'23, Denver, CO, USA	2023
▪ Program Committee (PC) Member, ISSRE'23 Artifact Evaluation (AE)	2022